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Palaeomagnetic Studies of Miocene Ignimbrites from Nevada

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Summary

A detailed palaeomagnetic study of a single ignimbrite shows marked variations in intensity, susceptibility, Koenigsberger ratio Q, and direction of magnetization. The behaviour of the intensity and Q can in part be explained as a change in grain size. A second profile through part of the same flow agrees well in its scalar properties, but the directions are very different. Six to fourteen samples have been collected from each of six additional ignimbrites in the same sequence. Their magnetic stability has been established by a.c. demagnetization and by lateral consistency over a distance of 60 km. Of all flows four are normally magnetized while the remainder have reversed polarity. The palaeomagnetic poles of two of these ignimbrites lie at 33°N and 49°N, the others lie within 20° of the rotational pole. Six of these ignimbrites have K–Ar ages between 29·7 and 17·8 My.

1. Introduction

The Tertiary ignimbrites of eastern Nevada and western Utah cover an area of many thousands of square miles (Fig. 1). A detailed description of the ignimbrites of Utah is given by Mackin (1960), of those in Nevada by Cook (1965) and Dolgoff (1963). The Nevada ignimbrites are in the rhyolite-dacite composition range and are welded to various degrees. The main minerals are quartz, sanidine, plagioclase, hornblende, and biotite but these rarely comprise more than 30 per cent of the rock, the remainder being glass. The lateral lithologic variation within units is usually small whereas the lithologic differences between units are great. The thicknesses range from 10 to about 1000 feet, the average being around 200 feet.

The ages used in this paper are those reported by Armstrong (1969). They were determined by the K–Ar method from separates of various minerals as well as from glass and whole rock samples. Multiple samples from widely scattered localities narrow the uncertainty to less than 0·5 My. The only undated ignimbrite in this study, the Petroglyph Cliff ignimbrite, lies stratigraphically between the Needles Range Formation and the Leach Canyon Tuff, which are 29·7 My. and 24·0 My respectively.

2. Vertical variations in a single ignimbrite

2.1.1 *Description*

The outcrop of the Swett and the base of the Bauers ignimbrite was sampled in detail in order to investigate vertical variations within a single flow. These samples were

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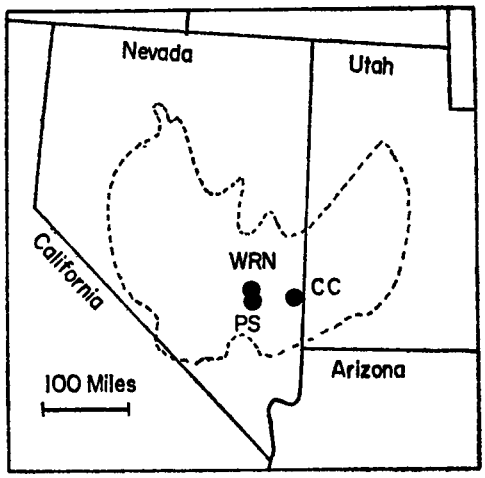


FIG. 1. Sampling locations. CC = Condor Canyon, WRN = White River Narrows, PS = Pahrock Springs. Dashed line indicates known extent of ignimbrite province (after Cook 1965).

drilled at about one metre stratigraphic intervals and were oriented with a suncompass. The collecting site lies in Condor Canyon, eastern Nevada (114·4° W 37·8° N, Fig. 1).

The Swett ignimbrite consists of five members which total 56 metres in thickness. The lowest member is black and glassy and is separated from the overlying members by a very irregular contact zone about half a metre thick. The other members are pink to pale purple in colour and are lithologically very similar except that the fourth member from the base contains many vesicules and the uppermost member although poorly exposed is structureless. The contacts between these members are sharp yet show no evidence of chilling although it is generally believed that each member represents a separate flow. All samples of the Swett ignimbrite are reversely magnetized.

2.1.2. *Magnetic mineral content*

Ten representative specimens were polished and examined by Mr H. R. Spall, using a Reichert Zeto-pan Pol microscope at 1250 magnification under oil immersion. The analysis revealed that the magnetic minerals, mainly titanohematite and some titanomagnetite and ilmenite, are in a highly oxidized state. Three main grain size ranges are apparent in all specimens; grains much larger than 100 microns, grains with dimensions from 5 microns to 100 microns, and grains from 5 microns down into the submicroscopic range. In numerical terms the last group comprises more than 60 per cent of the opaque minerals and occurs typically as lathes with a length to breadth ratio of 2:1 to 3:1. They are distributed randomly throughout the rock, and do not parallel the flow lineations visible macroscopically. Although the very small grains are numerous throughout the profile they are specially common near the base of the section. They are thought to be single or pseudo-single domain particles and it is considered that they account for the greater part of the remanent magnetization. Petrographically distinct titanohematite apparently of secondary origin is observed as irregular red patches along the margins and as needles parallel to the cleavage planes of biotite.

The Mössbauer effect (see e.g. Wertheim 1964) was exploited as an independent means of identifying the magnetic minerals, in particular those in the submicroscopic

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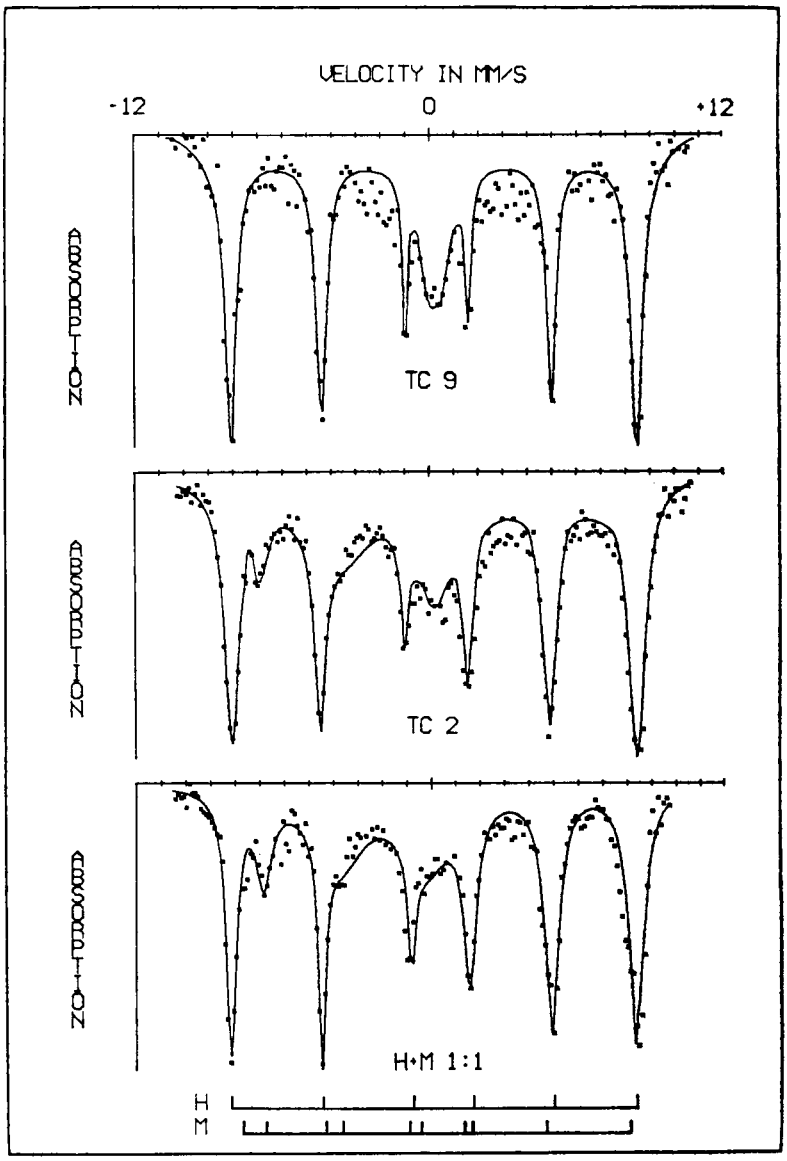


FIG. 2. Mössbauer spectra of ^{57}Fe of two ignimbrite samples and an artificial sample. Dots are original data points, solid line is a least square fit for the outer lines only. At the bottom of the figure the line positions of hematite (H) and magnetite (M) are indicated.

range. The spectra shown in Fig. 2 are obtained from the resonant absorbtion of ^{57}Fe. A mineral that is magnetically ordered yields a six line Zeemann pattern. Hematite is an example. If the iron atoms are located at two different lattice sites a double Zeemann pattern will result. At room temperature magnetite exhibits such a spectrum originating from Fe atoms at the A and B sites. On the other hand, paramagnetic minerals like hornblende or biotite give rise to a two line spectrum. Fig. 2 compares two Mössbauer spectra from the Swett ignimbrite with an artificial sample consisting of 50 per cent hematite and 50 per cent magnetite. At the bottom of the

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artificial sample a line drawing indicates the positions of the hematite and magnetite peaks. The centre portion of the rock spectra are rather complex due to the presence of paramagnetic minerals (biotite?, hornblende?). But since we are interested in the magnetic constituents only, no attempt was made to identify them and this part of the spectra has been excluded from further analysis.

Comparison with the artificial sample clearly shows that the basal sample (TC 2) contains both hematite *and* magnetite. Computer analysing the data reveals that they are present in a ratio of about 1:0·8. On the other hand, the spectrum TC 9 which is representative for all samples of the upper members shows hematite to be the only magnetic constituent. It should be pointed out, though, that small amounts of magnetite, say less than about 10 per cent of the magnetic fraction, would be difficult to detect by this method without having the sample sufficiently enriched by some separation technique.

The high percentage of magnetite in the basal sample and only in this sample is surprising. The very fact that it could not be seen with the microscope (1250 magnification) implies directly that the magnetite grains are smaller than a few microns.

In order to substantiate the validity of the interpretation of the Mössbauer spectra and to identify the carrier of the remanence four samples were stepwise demagnetized up to 2800 Oe, another four stepwise heated up to 725 °C (Fig. 3). Both techniques verify the different nature of the basal member (sample 1 and 2). Whereas the samples of the upper members retain more than 50 per cent of their natural remanence at

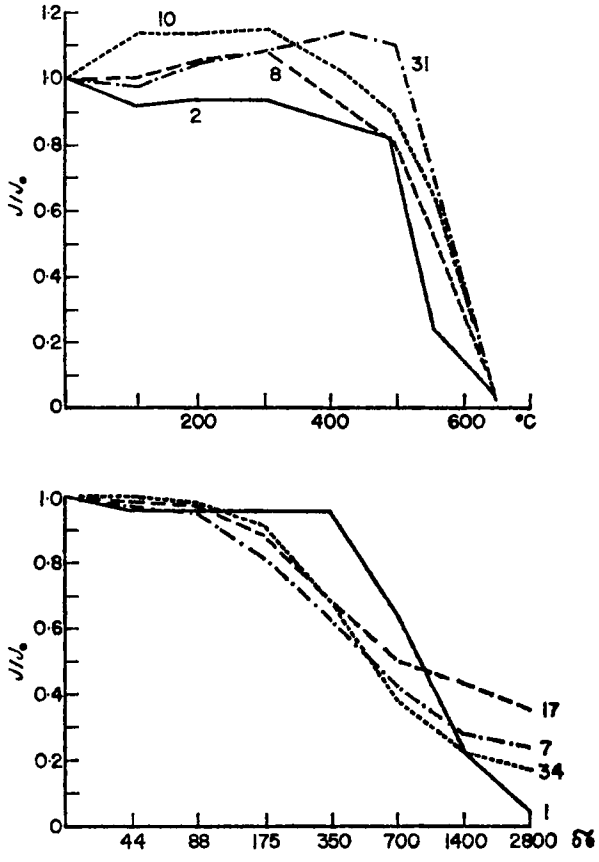


FIG. 3. Normalized thermal and a.c. demagnetization curves. Samples 1 and 2 are from the basal member, the others from the overlying members.

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560 °C, sample 2 has decreased to about 20 per cent. This implies that magnetite is the main carrier of the remanence in the basal member while it is hematite in the upper ones. The broad coercive force spectrum of the samples from the upper members ranging from 100 Oe to more than 2800 Oe indicates hematite of different grain sizes and titanium content. The basal sample, on the other hand, shows a coercive force spectrum consisting of two populations. The sudden decrease in intensity between 350 Oe and 1400 Oe results from the randomization of magnetite grains less than about 1 micron in diameter (Parry 1965). The high absolute intensity of $4{\cdot}4 \times 10^{-3}$ emu g^{-1} still retained at 2800 Oe necessitates the presence of fine grained hematite.

2.1.3. *Intensity and susceptibility*

Fig. 4 displays the vertical variation of some magnetic properties. The intensity of magnetization of the glassy base is the highest so far reported for ignimbrites. The magnetic moment of a standard rock-sample (a cylinder 2·5 cm in diameter and 2·5 cm long) is 2·5 emu. Remagnetization by lightning seems unlikely because these samples were collected at the bottom of a steep cliff and equivalent samples from an outcrop at White River Narrows and corresponding samples from the Bauers ignimbrite are extremely highly magnetized as well (0·3 emu and 1·3 emu respectively). In addition, one would expect larger variations in the direction of magnetization if lightning were the cause (Graham 1961). Yet, the high intensity of the two basal samples and the gradual decrease in intensity up to 27 metres above the base can readily be explained as a function of grain size. The microscopic studies revealed that grains at the limit of the visible range (a few microns) are numerous throughout the section but particularly so towards the base. If a Gaussian distribution is assumed then this implies an abundance of particles in the single or pseudo-single domain range (Larson *et al.*, 1969). Parry (1965) has shown that the thermoremanent magnetization of dispersed magnetite powder increases more than tenfold as the grain size decreases from 20 microns to 1·5 microns. Assuming a 3 per cent concentration of 1·5 micron particles in a volume equivalent to the samples used in this study the magnetic moment would

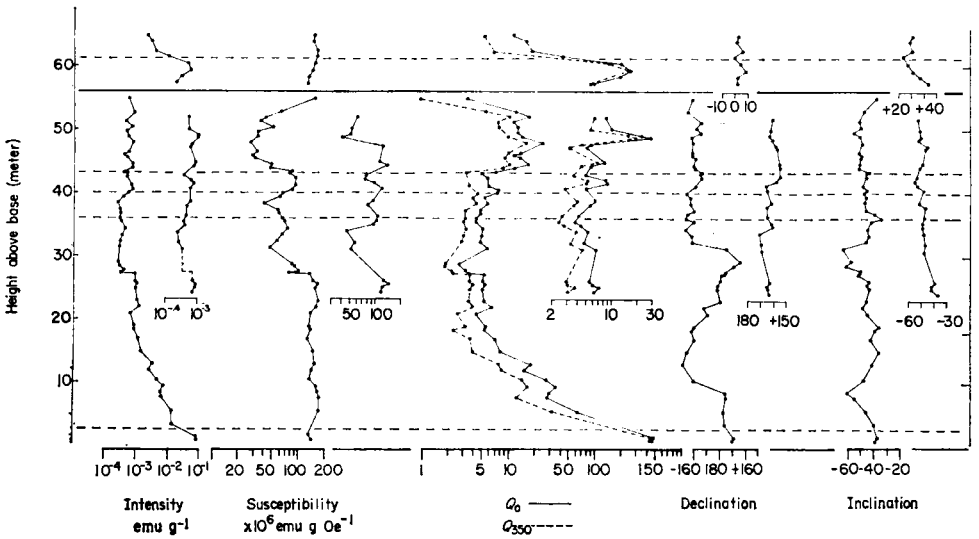


FIG. 4. Vertical variations in the Swett and basal part of the Bauers ignimbrites (profile 1 from 0 to 65 metres, profile 2 from 24 to 52 metres). Dashed horizontal lines represent contacts within the Swett ignimbrite, solid horizontal line marks the base of the Bauers ignimbrite.

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be 0·25 emu. In order to evaluate the changes in intensity as the grain size decreases further use can be made of the theoretical considerations by Stacey (1962). According to his calculations the magnetic moment per unit volume increases by another factor of ten as the grain size changes from 1·5 microns to 0·15 microns. The extremely high intensity in the glassy base can therefore be explained by postulating a grain size of the order of 0·1 micron. The decrease in intensity can then be accounted for as reflecting a decrease in the number of single domain particles. This interpretation is in agreement with the cooling history of an igneous body (see also the discussion of the Koenigsberger ratio below).

The above conclusion is further substantiated by susceptibility measurements (Figs 4 and 5). Although the susceptibility decreases with grain size (Nagata 1961, p. 28) this effect is small. Parry (1965) measured a decrease of the susceptibility of magnetite of about 15 per cent between grains greater than 20 micron and 1·5 micron. It is therefore safe to say that, in the present case, the susceptibility is proportional only to the amount of magnetic material within the experimental error. The relatively constant value in the basal 27 metres then eliminates variations in intensity as a function of concentration.

This is shown in a different way in Fig. 5 which demonstrates that there are two populations. For intensities higher than about 10^{-3} emu g^{-1} the susceptibility remains constant at the maximum value for this ignimbrite. The second group shows no correlation between the two quantities.

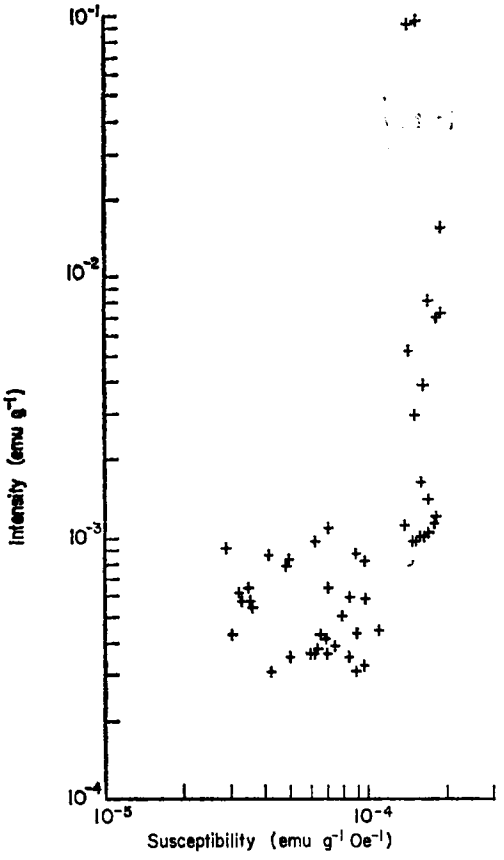


FIG. 5. Relationship between intensity and susceptibility for the Swett ignimbrite, profile 1.

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The sudden decrease of the intensity as well as of the susceptibility at 27 metres above the base is surprising in that it does not coincide with any visible flow boundary. Above this level the susceptibility follows no clear trend.

2.1.4 *The Koenigsberger ratio*

The Koenigsberger ratio Q is used here as the ratio of intensity to susceptibility, the subscripts 0 and 350 referring to the values measured before and after demagnetization in a 350 Oersted field. All Q values are very large, some reaching a value of several hundred (Fig. 4). Stacey (1967) has shown that Q increases for grain sizes less than about 20 microns. If some extrapolation is permissible then most of these Q values, excluding the very large ones, would correspond to particles between 0·1 and 1·0 micron which is in general agreement with the microscopic observations and the inferences drawn from the intensities.

Hatherton (1954) observed on ignimbrites from New Zealand a change from high Q values at both flow boundaries to low values in the interior of the flow. He demonstrated that the Q variations can be related to the cooling history of individual flow units yet he was unable to find a connection between the two. The faster cooling rate at the upper and lower surfaces of a flow results in smaller grain sizes and the variation of Q as reported by Hatherton (1954) therefore follows immediately from Stacey's (1967) work. Such an interpretation accounts well for the variations in the lower half of the Swett ignimbrite. Towards the top of the flow Q increases with exception of the uppermost two values. If one postulates a partial remagnetization by the overlying Bauers ignimbrite which is of opposite polarity then one would expect a decrease in intensity and the Koenigsberger ratio. The observed variations are therefore in accord with the cooling history of an igneous body.

2.1.5 *Direction of magnetization*

In contrast to the non-directional quantities the declination and inclination show no consistent relationship (Figs 4 and 6). AC demagnetization in a 350 Oe field did not cause any change. To test the stability further, four specimens were stepwise demagnetized up to 2800 Oe. The direction remained constant within 5°. Stepwise heating of another four specimens did not alter the directions either. Upon heating to 725 °C and cooling in a field of 0·5 Oe the samples reflected the applied field direction indicating that the susceptibility anisotropy, if present, is not sufficient to influence the direction of magnetization beyond the experimental error.

It is tempting to consider the change in direction at 32 metres above the base as a reheating effect. Heatflow calculations show that in order to reheat the top four metres of the lower member above the Curie point the upper members must have been emplaced only a few years later. At this time the centre portion of the lower member was still above its Curie point and should therefore have the same direction as the upper members. At the 95 per cent confidence level this is indeed the case.

2.2 *Local correlation*

A portion of the same section has been collected in a second profile about 100 metres away on the opposite side of the canyon (Fig. 4). The object was to see how persistent the data are over a short distance where the stratigraphic control is excellent and to investigate the possibility of using magnetic properties as an aid in regional correlation.

The scalar properties agree rather well, in particular the Koenigsberger ratio. But the directions, though much more consistent in themselves, differ markedly (see also Table I). The mean direction of this profile is statistically different at the 95 per cent

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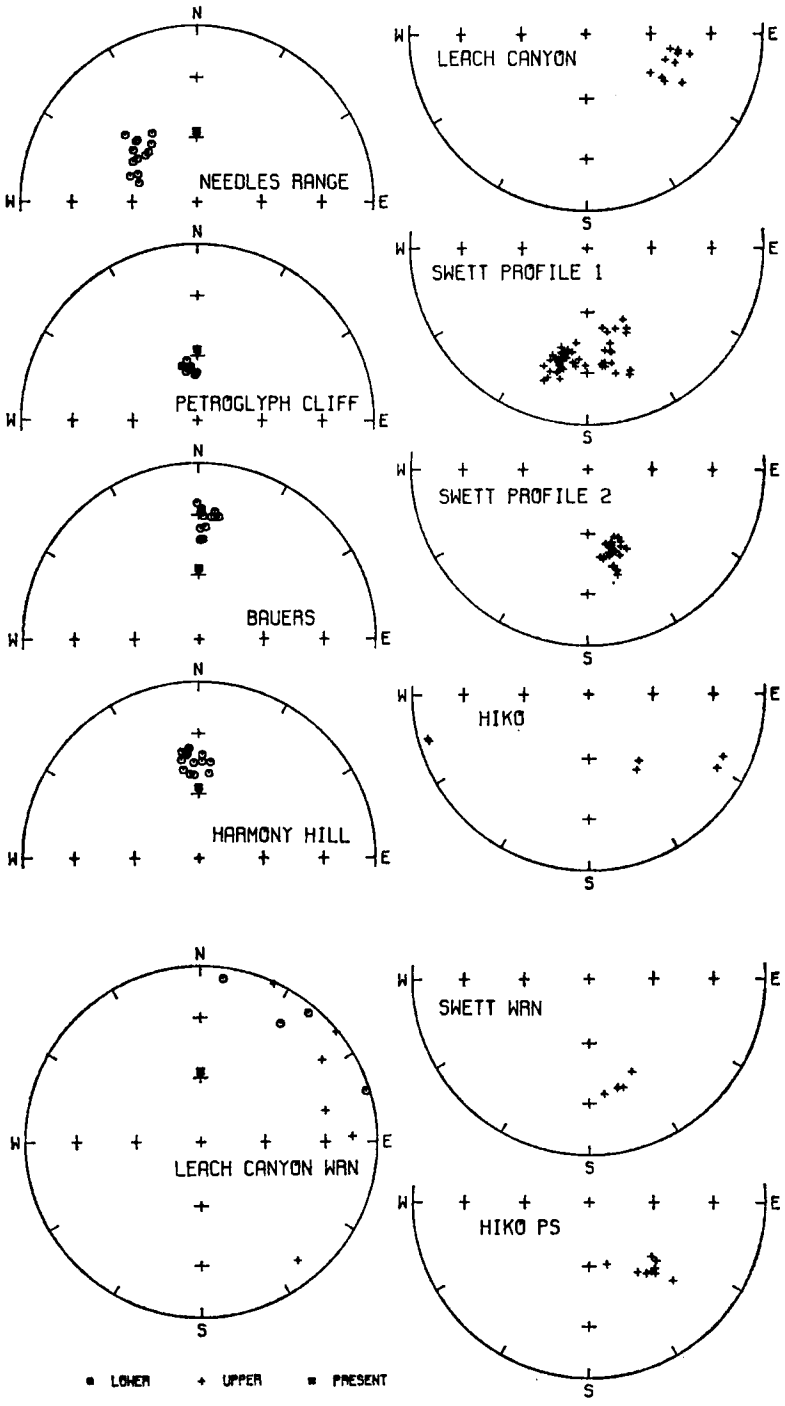


FIG. 6. NRM directions of ignimbrites from Condor Canyon, White River Narrows (WRN), and Pahrock Springs (PS). Equal area projection.

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confidence level from the corresponding part of the first profile. In the light of this fact it is questionable how much meaning can be attached to the heatflow model suggested by the first profile.

Doell & Cox (1963) showed that about six samples per lava flow are sufficient to warrant a good determination of the direction of magnetization. As the above data clearly show this would not have been sufficient for the Swett ignimbrite, i.e. a volcanic ash flow. Whether or not this may be generalized to all ash flows has to await further results from similar studies.

2.3 *The Bauers ignimbrite*

The Swett ignimbrite is overlain by the Bauers ignimbrite which has about four metres of black glass at its base. This basal unit is overlain by a red purple, highly welded flow of which the lowest four metres have been collected.

The Bauers ignimbrite is normally magnetized, nevertheless, the nondirectional properties undergo variations similar to the basal part of the Swett ignimbrite (Fig. 4). High intensities and large Q values are observed in the lowest unit and both quantities decrease upward yielding support for the inferences drawn from observations in the Swett ignimbrite. It appears that high intensities and large Q values are characteristic of the basal glassy unit of these ignimbrites.

3. Regional studies

Six to fourteen samples per flow have been collected from each of five additional ignimbrites at the same locality in Condor Canyon (Fig. 1). They are spaced within about 100m^2 and are magnetically oriented. This collection does not constitute an ideal population but it is sufficient to determine the polarity of a flow. The NRM directions of all flows from Condor Canyon are shown in Fig. 6.

Five of these ignimbrites were collected at a second locality some 60 km west of Condor Canyon. The structural control at these sites is somewhat uncertain so that perfect agreement with the Condor Canyon data cannot be expected. But since the tilt nowhere exceeds 20° such a correction would not change the interpretation significantly. In particular, it could not have any effect on the polarity of magnetization. All flows exhibit the same polarity as their counterparts in Condor Canyon. Three examples are shown in Fig. 6.

The grouping of the directions remains essentially the same after a.c. demagnetization in a 350 Oe field except for the Hiko Tuff from Condor Canyon and the Leach Canyon ignimbrite from White River Narrows. Both of these units exhibit large scatter in the initial NRM direction and this is greatly reduced upon demagnetization. It is interesting to note that the dispersion of NRM directions is not a characteristic of a certain ignimbrite but rather of the sampling locality, an observation similar to one made by Cox (1961) on basalts from Idaho. The data of all flows from Condor Canyon are summarized in Table 1.

Out of the seven units four are normally magnetized (Harmony Hills, Petroglyph Cliff, Needles Range), three are reversed (Hiko, Swett, Leach Canyon). The palaeomagnetic poles of the Needles Range and Leach Canyon ignimbrites are far removed from the expected Tertiary pole. The possibility of an erroneous bedding correction is definitely excluded since all these flows in Condor Canyon belong to one structural unit. The remaining five ignimbrites yield pole positions within 20° of the rotational pole. Applying Fisher statistics results in a mean pole with the co-ordinates 109·4° E, 80·6° N. The 95 per cent cone of confidence has a semi-angle $\alpha_{95} = 12{\cdot}7°$, the precision parameter is $k = 37$.

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Table I

Summary of magnetic data and Fisher statistics from Condor Canyon. For NRM data N includes all specimens measured. (Decl = declination, Incl = inclination, N = number of samples, R = length of resultant vector, k = precision parameter, α_{95} = semi-angle of 95 per cent cone of confidence, Long = longitude of palaeomagnetic pole, positive to the west, Lat = latitude. Note: all poles have been transferred to the Northern Hemisphere.

	Age in My.		Decl	Incl	N	R	k	α_{59}	Long.	Lat.
Hiko	17·8±0·2	NRM	108·1	−22·4	6	4·95	5	34·4		
		AC 350	166·2	−49·2	6	5·97	189	4·9	−125·4	76·2
Harmony Hills	21·0±0·2	NRM	−3·5	44·0	14	13·93	175	3·0	−80·3	77·6
		AC 350	−17·3	51·0	10	9·96	229	3·2	−137·4	74·5
Bauers	21·5±0·2	NRM	3·5	33·2	13	12·91	135	3·6	−55·9	70·1
		AC 350	3·5	33·3	6	5·96	157	5·4	−55·9	70·2
Swett, profile 1	23·3±0·2	NRM	−176·3	−37·1	53	51·70	40	3·1	−53·9	72·6
		AC 350	−177·2	−37·6	46	44·75	36	3·6	−56·6	73·1
Swett, profile 2		NRM	163·2	−49·3	23	22·89	203	2·1	−131·8	74·1
		AC 350	163·4	−51·0	23	22·92	297	1·7	−136·4	75·1
Swett, combined		NRM	178·3	−41·2	76	73·70	33	2·9	−72·0	75·7
		AC 350	177·0	−42·6	70	67·79	31	3·1	−77·3	76·7
Leach Canyon	24·0±0·5	NRM	109·6	−47·2	10	9·92	108	4·7	−168·8	31·6
		AC 350	113·3	−43·2	6	5·96	129	5·9	−163·6	32·9
Petroglyph Cliff		NRM	−9·7	65·6	9	8·99	727	1·9	146·6	77·7
		AC 350	−8·9	65·8	6	5·99	422	3·3	143·9	77·9
Needles Range	29·7±0·3	NRM	−50·6	55·1	13	12·88	98	4·2	−168·6	49·9
		AC 350	−53·4	57·4	10	9·95	193	3·5	−173·0	48·5

4. Conclusion

In the light of the variations observed in the Swett ignimbrite it is debatable how accurately these rocks reflect the geomagnetic field direction at the time of their deposition. But the directions of the Needles Range and Leach Canyon ignimbrites are clearly outside the range of any reasonable variations one might expect within a single flow. The possibility of remagnetization at a later time can be rejected since it could result in a direction only closer to the present pole.

'Anomalous' poles are fairly commonly observed in Tertiary rocks (e.g. Irving 1964, Grommé 1965). Most of these results are based on igneous rocks which cool below their Curie point in a very short time and therefore represent spot readings. The anomalous pole positions may thus be indicative of large secular variations and it is suggestive to speculate that this may reflect another characteristic of the great instability of the Earth's magnetic field during the Tertiary. A second possibility is that these rocks record an intermediate direction acquired during a magnetic polarity transition (Watkins 1969).

Acknowledgement

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